Nivaaran: External SOAP Partner Integration

Contract Specification, HTTP Binding, Discovery, and Fault Mapping

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Task 1: Partner Edge Selection and SOAP Justification

External Partner Selected: NationalSMS Gateway (National Telecom Infrastructure Ltd.)
Operation Integrated: sendSMS(recipientNumber, senderId, messageText, priority)

While all internal Nivaaran microservices (Accounts, Directory, Complaints, Resolution, Notifications) communicate over lightweight REST APIs with JSON payloads, the external SMS gateway at the enterprise boundary requires SOAP for three concrete architectural reasons:

1. **Rigid Schema and Contract Enforcement:** Telecom providers operate large-scale clearinghouses where every dispatch must strictly satisfy data validation before acceptance. XML Schema in WSDL enforces phone format validity, sender authentication, and message payload bounds deterministically before runtime execution.

2. **Message-Level Security (WS-Security):** In telecom networks, HTTP traffic passes through intermediate proxies, telco gateways, and regional message switches. Transport-layer TLS only secures point-to-point connections; WS-Security signs and encrypts the SOAP header end-to-end, guaranteeing non-repudiation and credential protection across intermediate routing hops.

3. **Transactional Delivery and Non-Repudiation:** High-volume SMS dispatches incur financial billing and regulatory audit obligations. SOAP message exchanges guarantee delivery receipts with immutable message identifiers, enabling legally binding audit trails for campus emergency grievances.

Task 4: HTTP Request Binding

The SOAP request is carried over HTTP/1.1 as a POST operation to the endpoint specified in the WSDL's service/port element. The request headers specify the XML content type and the SOAPAction header matching the WSDL binding.

```
POST /ws/sms HTTP/1.1
Host: api.telecomsms.internal
Content-Type: text/xml; charset=utf-8
Content-Length: 1042
SOAPAction: "urn:nivaaran:telecom:sms/sendSMS"

<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
               xmlns:sms="urn:nivaaran:telecom:sms"
               xmlns:wsse="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1
```

```
  <soap:Header>
    <wsse:Security soap:mustUnderstand="1">
      <wsse:UsernameToken>
        <wsse:Username>nivaaran_notifications_client</wsse:Username>
        <wsse:Password Type="...#PasswordDigest">dGhpcy1pcy1hLXNlY3VyZS1kaWdlc3Q=</wsse:Password>
      </wsse:UsernameToken>
    </wsse:Security>
  </soap:Header>
  <soap:Body>
    <sms:sendSMSRequest>
      <sms:recipientNumber>+919876543210</sms:recipientNumber>
      <sms:senderId>NIVAARAN</sms:senderId>
      <sms:messageText>Your grievance CMP-2026-1042 has been assigned.</sms:messageText>
      <sms:priority>HIGH</sms:priority>
    </sms:sendSMSRequest>
  </soap:Body>
</soap:Envelope>
```

Task 5: Modern Discovery and Registry Entry

In modern cloud architectures, public UDDI operator registries have been superseded by direct URL configuration and managed API catalogues. For Nivaaran, the telecom partner provides a published entry in the campus enterprise service catalogue.

Two Discovery Mechanisms in Practice:

• **Direct URL Configuration:** The telecom provider issues their production WSDL URL directly (https://telecomsms.internal/contracts/sms-v1.wsdl), which Nivaaran's deployment loads via an environment variable (SMS_GATEWAY_WSDL). This is the standard operational pattern.

• **Internal Service Catalogue (Modern UDDI):** An internal enterprise service directory allows service discovery via metadata lookup.

Catalogue Field	Registry Value / Description
businessEntity	National Telecom Infrastructure Ltd. (Partner ID: TEL-IN-9081)
businessService	NationalSMS Gateway (Automated Mobile Alerts Service)
bindingTemplate	SOAP 1.1 Document/Literal over HTTPS on Port 443
accessPoint (URL)	https://api.telecomsms.internal/ws/sms
tModel (Spec Pointer)	urn:nivaaran:telecom:sms:v1 (Points to partner.wsdl schema specification)

Task 6: Fault Mapping and Boundary Isolation

A foundational microservice principle is that *the partner's vocabulary must never leak into internal domain boundaries or caller responses*. When the external SOAP gateway returns a soap:Fault, Nivaaran's Notifications service translates it into the system's standard RFC 7807 Problem Details representation.

External SOAP Partner Fault	Nivaaran Internal Mapping	Published Response to Caller
faultcode: soap:Client **faultstring:** Invalid Mobile Number **detail:** INVALID_MSISDN	Notifications service intercepts SOAP fault, logs audit failure, and generates standard domain error.	**HTTP Status:** 422 Unprocessable **type:** /errors/invalid-recipient **title:** Invalid Recipient Contact **detail:** The provided mobile contact is invalid.
faultcode: soap:Server **faultstring:** SMS Gateway Overloaded **detail:** QUEUE_EXHAUSTED	Notifications service stores alert in local outbox table for retry, maintaining zero lost complaints.	**HTTP Status:** 202 Accepted **status:** alert_queued Caller is isolated from partner disruption.